

Windows shim (wireguard-nt + service)

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Volume 4 (Clients) nano-detail specification — deepens the **Windows row** of the per-platform shim matrix in the pass-1 client overview [04_UI §5/§6, 03-client §5.3]. Where doc 03 says *one sentence* ("a privileged Windows service hosts the core and the wireguard-nt adapter; the Flutter app talks to it over a named pipe"), this document pins the **process model, the named-pipe IPC wire protocol, the SYSTEM service lifecycle, the wintun/wireguard-nt adapter handoff, WFP-based split-tunnel + kill-switch, the Authenticode signing chain, and the TunnelPlatform contract implementation** to near-code granularity.

SPEC ONLY — it describes *what to build and why* (signatures, state machines, wire formats, error taxonomy, memory budgets, edge cases, test points). It does not build the product. Sources cited inline by id: [03-client §N] = final/03-client-core-and-ui.md; [04_UI §N] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [01-orch §N] = final/v02-data-plane/orchestrator-and-state.md; [01-trans §N] = final/v02-data-plane/transport-trait.md; [SYN §N] = v09-research/_SYNTHESIS.md. External Windows-platform facts that the evidence base does not itself contain are tagged UNVERIFIED per constitution §11.4.6 — recommendations to be ratified against the Microsoft WFP / Service / wintun / wireguard-nt documentation and a real machine, never fabricated.

Boundary with sibling docs. This document **consumes** the FFI surface + TunnelStatus enum owned by [03-client §3] and the orchestrator/ TunnelStatus contract owned by [01-orch §4.1], and the TunnelPlatform channel contract owned by [03-client §4]. It **owns** everything Windows-specific from those contracts down to the OS: the service, the pipe, the adapter, WFP, and signing. It does **not** redefine the FFI surface, the Transport trait [01-trans], the orchestrator loops [01-orch §3], nor the WG crypto core.

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0. Position, ownership, and the Windows-specific invariants

Windows is the one desktop target where the tunnel **cannot** run in the same process as the Flutter UI: creating a TUN adapter, loading a driver, and installing WFP filters all require administrator / SYSTEM privilege, while the Flutter app runs unprivileged in the interactive user's session [04_UI §6, 03-client §5.3]. The shim is therefore split across a **privilege boundary**: a long-lived **privileged Windows service** (HelixTunnelSvc, runs as LocalSystem) hosts helix-core + the network adapter, and the unprivileged Flutter app drives it over a **named pipe** that satisfies the TunnelPlatform contract [03-client §4].

This document owns five Windows-specific contracts and nothing else:

#	Contract	Owned here	Consumed from
W-C1	Process/trust model — unprivileged UI $\rightleftharpoons$ SYSTEM service split; who holds the adapter, WFP, and the core	§1	OS privilege model
W-C2	Named-pipe IPC protocol — pipe name, security descriptor, framing, the request/response + event envelope that carries the FFI surface [03-client §3] and TunnelPlatform verbs [03-client §4]	§4	[03-client §3/§4]
W-C3	Adapter handoff — wintun (production) / wireguard-nt (optional plain-udp fast path); how the service pumps packets into helix-core without a Unix fd	§2–3	wintun / wireguard-nt API
W-C4	WFP kill-switch + split-tunnel — the Windows realisation of the core's kill-switch/DNS state effects [01-orch §8] and per-app/per-route split [03-client §4 TunnelConfig]	§8	[01-orch §8]
W-C5		§9	operator-supplied certs

#	Contract	Owned here	Consumed from
	Signing chain — Authenticode for the service + installer, driver trust for wintun/wireguard-nt (§11.4.133)		

0.1 Windows-specific invariants (extend, never weaken, the client invariants)

#	Invariant	Source / rationale
W-I1	The unprivileged UI never touches the adapter, the driver, or WFP. All privileged operations happen only inside HelixTunnelSvc; the UI's only capability is "send a TunnelPlatform command, receive events" over the pipe. A compromised UI cannot install firewall rules or open a TUN.	least privilege; W-C1
W-I2	helix-core is the <i>only</i> place WG + transport live — the service hosts the same Rust core as every other platform; it does not re-implement crypto or obfuscation. wintun is a <i>generic L3 carrier</i> , not a WG implementation (O2 [03-client §4], I4 [01-trans §1]).	one core, no fork
W-I3	Kill-switch filters survive a service crash (fail-closed). WFP kill-switch filters are installed in a persistent sub-layer so an abnormal service exit leaves egress blocked (O-I11 [01-orch §8]); only a <i>graceful user stop</i> removes them. Split-tunnel exclusion filters are dynamic (auto-removed when the engine handle closes with the tunnel).	O-I11; §8.4
W-I4	The pipe is authenticated by its DACL, not by trust in the caller. The pipe's security descriptor restricts CONNECT to the interactive desktop user(s); the service verifies the connecting client's token (§4.3) before honouring any command.	W-C2; §11.4.10
W-I5	Status truth flows core → service → pipe → UI; the UI never paints "connected" on intent. The pipe re-emits the core's TunnelStatus [01-orch §4.1] byte-for-byte; the UI is a pure function of that stream (CI2 [03-client §0.1]).	CI2 honesty
W-I6	No secret crosses the pipe in cleartext logs or tracked files. The session/map token + any transport SecretBytes [01-trans §3.2] ride the pipe body but are never logged; the pipe body is local-only (kernel-mediated), never persisted (§11.4.10).	§11.4.10

flowchart TB

```

    subgraph user["Interactive user session (Medium IL, unprivileged)"]
        UI["Flutter app (helix-ui)\nWindowsTunnelPlatform : TunnelPlatform"]
        PC["named-pipe CLIENT\n(dart:ffi → kernel32 CreateFile/Read/Write)"]
        UI --> PC
    end
  
```

```

    subgraph sys["HelixTunnelSvc – Windows service (LocalSystem,
SYSTEM IL)"]
        PS["named-pipe SERVER\n\\\\.\\pipe\\helixvpn\\tunnel"]
        CORE["helix-core (Rust, in-service)\norchestrator + helix-
wg + helix-transport"]
        ADP["wintun adapter (generic L3 TUN)\n(opt: wireguard-nt
for plain-udp)"]
        WFP["WFP engine\nkill-switch (persistent) + split-tunnel
(dynamic)"]
        PS <--> CORE
        CORE <--> ADP
        CORE --> WFP
    end
    PC <== "named-pipe IPC (length-prefixed frames)" ==> PS
    ADP -. "encrypted/obfuscated datagrams" .-> NET[["gateway
:443 / :51820"]]

```

1. Process & trust model (why a SYSTEM service at all)

Three facts force the split [04_UI §6, 03-client §5.3]:

1. **wintun / wireguard-nt require privilege.** Creating or opening a tunnel adapter and loading the kernel driver need `SeLoadDriverPrivilege` + administrator rights; WFP filter installation needs `FWPM_IOCTL_ADD` (administrator). The interactive Flutter app does not and must not hold these.
2. **A tunnel must outlive the UI.** The user closes the app window; the VPN stays up. A service is the OS-native "runs without an interactive session, survives logoff" host.
3. **One privileged surface, audited once.** Concentrating every privileged operation in one signed service (W-I1) gives a single, small, reviewable attack surface instead of scattering elevation prompts through the UI.

1.1 Who runs as what

Component	Process	Security context	Holds
Flutter UI + WindowsTunnelPlatform	helix_access.exe	interactive user, Medium IL	nothing privileged — only the pipe client
HelixTunnelSvc	helix_tunnel_svc.exe	LocalSystem (SYSTEM IL)	the core, the adapter, WFP filters, the pipe server

Component	Process	Security context	Holds
wintun / wireguard-nt	kernel driver	kernel	the L3 adapter object

Service-account hardening (recommended, §11.4.133). LocalSystem is the simplest account that can load the driver + edit WFP. A hardened variant runs the service under a dedicated **virtual service account** (NT SERVICE\HelixTunnelSvc) granted only the required privileges (SeLoadDriverPrivilege, the WFP FWPM_IOCTL_* rights, SeChangeNotifyPrivilege) and a write-restricted token. UNVERIFIED: whether wintun adapter creation succeeds under a non-LocalSystem service SID is a Phase-1 SEC/IT measurement, not an assumption — surfaced as decision **D-WIN-3** (§13).

2. The adapter: wintun vs wireguard-nt (the load-bearing choice)

The doc-03 row reads "wireguard-nt/wintun" [03-client §5.3]; this section pins **which one carries the obfuscated datapath and why**, honouring W-I2 (one core, no fork).

Adapter	What it is	Carries	Compatible with helix-core's obfuscation ladder?
wintun	a generic, signed L3 TUN adapter (raw IP packets to/from userspace via a ring buffer)	whatever userspace puts on the ring	Yes — the service does WG + Transport in helix-core and pumps the wintun ring, exactly mirroring the iOS packetFlow ≠ core pump [03-client §5.1]
wireguard-nt	the kernel WireGuard driver — does WG Noise + ChaCha20-Poly1305 in kernel , speaks plain WG-over-UDP only	inner WG only, plain UDP	No for masque-h3/shadowsocks/lwo/... — kernel WG cannot be wrapped in the helix-transport carriers; using it would duplicate WG outside the one core (violates O2/I4)

Decision (surfaced as D-WIN-1, §13). The **production carrier is wintun**: the service runs helix-core (WG + the full Transport ladder [01-trans §6]) in userspace and pumps the wintun ring, so every obfuscation transport works identically to every other platform (W-I2, I4 [01-trans §1]). **wireguard-nt is an optional kernel fast path for the plain-udp rung only** — when the ladder selects plain-udp on a cooperative network it MAY hand the WG config to wireguard-nt for in-kernel throughput, and MUST fall back to the wintun+core path the instant the ladder escalates to any obfuscated rung. Phase-1 ships **wintun only**; the wireguard-nt fast path is a Phase-2 throughput optimisation gated on a BENCH measurement (§12).

2.1 wintun adapter handle & session (the packet ring)

The wintun API (wintun.dll, the official WireGuard component) exposes an adapter + a session holding a ring buffer; userspace allocates/sends and receives/releases packets. The helix-core Windows shim wraps it behind a WinTunDevice that satisfies the same packet-source/sink role the Unix helix-tun::TunDevice plays [01-orch §2.1].

```
// helix-core/src/platform/windows/wintun.rs (the Windows
    TunDevice impl)
// FFI to wintun.dll – names per the official WireGuard wintun.h.
    UNVERIFIED: exact
// argument list pinned against the shipped wintun.h header, not
    memory.
#[link(name = "wintun")]
extern "C" {
    fn WintunCreateAdapter(name: PCWSTR, tunnel_type: PCWSTR,
        guid: *const GUID) -> WINTUN_ADAPTER_HANDLE;
    fn WintunStartSession(adapter: WINTUN_ADAPTER_HANDLE,
        capacity: u32) -> WINTUN_SESSION_HANDLE; // capacity =
        pow2, 0x200000..=0x4000000
    fn WintunGetReadWaitEvent(session: WINTUN_SESSION_HANDLE) ->
        HANDLE; // wait-handle for rx
        readiness
    fn WintunReceivePacket(session: WINTUN_SESSION_HANDLE, size:
        *mut u32) -> *mut
        u8; // borrow an inbound IP packet
    fn WintunReleaseReceivePacket(session: WINTUN_SESSION_HANDLE,
        packet: *const u8);
    fn WintunAllocateSendPacket(session: WINTUN_SESSION_HANDLE,
        size: u32) -> *mut u8; // borrow an outbound
        slot
    fn WintunSendPacket(session: WINTUN_SESSION_HANDLE, packet:
        *const u8);
    fn WintunEndSession(session: WINTUN_SESSION_HANDLE);
    fn WintunCloseAdapter(adapter: WINTUN_ADAPTER_HANDLE);
}

/// Windows L3 carrier – drop-in for the orchestrator's TunDevice
    role [01-orch §2.1].
pub struct WinTunDevice {
    adapter: WintunAdapter, // RAII: WintunCloseAdapter on Drop
    session: WintunSession, // RAII: WintunEndSession on Drop
    read_evt: HANDLE, // WintunGetReadWaitEvent – awaited
        on the rx loop
}

impl WinTunDevice {
    /// Create+configure the adapter: overlay IP, routes
        (AllowedIPs), MTU, tunnel DNS.
    /// IP/route/DNS are set via the iphlapi / netsh route table,
        NOT via wintun (it is L3-only).
    pub fn create(cfg: &TunnelConfig) -> Result<Self,
        WinShimError>; // §4.5 TunnelConfig
```

```

    /// Cancel-safe async recv: wait on read_evt,
    /// WintunReceivePacket → copy out → release.
    pub async fn recv(&self) -> Result<Bytes,
    WinShimError>;           // → orchestrator loop A [01-
    orch §3.1]
    /// Send a decrypted inbound IP packet onto the ring
    /// (orchestrator loop B) [01-orch §3.2].
    pub async fn send(&self, ip_pkt: Bytes) -> Result<(),
    WinShimError>;
}

```

The orchestrator's three loops [01-orch §3] are **unchanged** on Windows: loop A reads `WinTunDevice::recv` (plaintext IP) → WG encrypt → `Transport::send`; loop B `Transport::recv` → WG decrypt → `WinTunDevice::send`. The wintun ring replaces the Unix `tun fd`; nothing else in the core differs (W-I2). IP address, routes (AllowedIPs), and tunnel DNS are applied via `iphlpapi` (`CreateUnicastIpAddressEntry`, `CreateIpForwardEntry2`, `SetInterfaceDnsSettings`) against the adapter's LUID — not via wintun, which is pure L3 transport.

3. The in-service core host (FFI surface on Windows)

The service hosts the **same Rust FFI surface** authored in `helix-ffi/src/api.rs` [03-client §3.1] — but unlike Dart-driven platforms, on Windows the *consumer* of that surface is the **pipe server inside the same process**, not `flutter_rust_bridge`. The FFI surface is therefore called natively (Rust → Rust), and its `StreamSink<TunnelStatus>` is wired to the pipe's event pump (§4.4) rather than to `frb`.

3.1 The surface the service drives (consistent with [03-client §3.1] and [01-orch §4.1])

```

// The service calls these exact functions; the TunnelStatus they
// emit is the CANONICAL
// orchestrator enum [01-orch §4.1], re-emitted over the pipe
// verbatim (W-I5).
pub async fn start(cfg: ClientConfig) ->
    anyhow::Result<()>;           // ClientConfig [03-client §3.1]
pub async fn stop() -> anyhow::Result<()>;
pub fn status_stream(sink:
    StreamSink<TunnelStatus>);           // → pipe event
    pump (§4.4)
pub async fn exits() -> anyhow::Result<Vec<ExitOption>>;
pub async fn set_exit(id: String, multi_hop_chain:
    Option<Vec<String>>) -> anyhow::Result<()>;
pub async fn set_shields(s: Shields) -> anyhow::Result<()>;
pub async fn advertise(cidrs: Vec<String>) ->
    anyhow::Result<AdvertiseResult>; // connector flavor

```

3.2 The Windows TUN handoff — `attach_tun(fd)` has no fd here

The cross-platform FFI exposes `attach_tun(fd: i32)` for Unix fd platforms (Android `ParcelFileDescriptor`, Linux `tun fd`) [03-client §3.1]. **Windows has no fd**: wintun hands userspace a *ring*, exactly as iOS hands the extension a `packetFlow` [03-client §5.1]. The core therefore exposes the **pump-callback variant** (identical in shape to the iOS `helix_core_set_tun_writer / helix_core_tun_out` pair [03-client §5.1]), and the Windows service wires `WinTunDevice` into it:

```
// helix-ffi/src/api.rs – the writer-callback pump variant
// (Windows + iOS; consistent surface)
pub fn attach_tun_pump(reader: WinTunReader, writer:
    WinTunWriter) -> anyhow::Result<()>;
pub fn detach_tun() ->
    anyhow::Result<()>; // shared with the fd variant [03-
    client §3.1]
```

Consistency note (§11.4.6). `attach_tun(fd)` and `attach_tun_pump(reader, writer)` are the two members of *one* TUN-handoff contract: fd-platforms call the former, ring/packetFlow platforms (Windows, iOS) call the latter. The exported set, the `TunnelStatus` enum, and the `Shields/ExitOption/AdvertiseResult` types are otherwise **byte-for-byte the doc-03 surface** [03-client §3.1] — the FFI contract test (§12 UT) asserts the mirror.

3.3 The canonical `TunnelStatus` the pipe carries (CONSISTENT with [01-orch §4.1])

The service marshals the **orchestrator's** five-variant `TunnelStatus` [01-orch §4.1] onto the pipe verbatim — this is the frozen core contract:

```
// helix-core/src/status.rs – the wire form on the pipe (W-I5)
// [01-orch §4.1]
#[derive(Clone, Debug, PartialEq, Eq)] // + serde for the pipe
// envelope (§4.2)
pub enum TunnelStatus {
    Connecting, // dialling a
    // rung
    Handshaking, // carrier up,
    // WG handshake in flight
    Connected { transport: String, rtt_ms: u32 }, // e.g.
    // ("masque-h3", 23)
    Reconnecting, // a working
    // tunnel dropped; re-dialling
    Down { reason: String }, // terminal for
    // this attempt (§11 reasons)
}
```

The **client-side FFI extension** (`Disconnected`, `Connected.path`, `Danger{kind}`) defined in [03-client §3.1] is layered **above** the pipe in the Dart `helix_core_ffi` package, derived from this core enum + `PlatformTunnelEvents`

(§5.2) — it is NOT a second wire enum. Keeping the pipe on the orchestrator's five-variant enum is what makes the Windows pipe consistent with [01-orch §4.1] while the UI still gets the richer doc-03 surface (mapping in §5.2).

4. Named-pipe IPC — the UI ⇔ service wire protocol

The named pipe is the single seam between the unprivileged UI and the privileged service. It is the Windows TunnelPlatform transport: every TunnelPlatform verb [03-client §4] and the core's status stream cross it.

4.1 Pipe identity & connection model

Property	Value	Rationale
Pipe name	\\.\pipe\helixvpn\tunnel	well-known, single instance; service is the server
Type	byte-stream, duplex (PIPE_ACCESS_DUPLEX), message-read mode (PIPE_READMODE_MESSAGE) UNVERIFIED (byte-mode + own framing is the fallback)	one channel carries both req/resp and pushed events
Instances	PIPE_UNLIMITED_INSTANCES capped at a small N (e.g. 4)	one UI + headroom; reject excess (§11)
Server	HelixTunnelSvc creates it at service start (CreateNamedPipeW)	the privileged side owns it
Client	the UI opens it (CreateFileW via dart:ffi, §5.1)	unprivileged side connects

4.2 Frame format & message envelope

Every message is a **4-byte little-endian length prefix + a serialized body**. Body encoding is a self-describing binary format (recommended **MessagePack** via rmp-serde; UNVERIFIED final pick — JSON acceptable for debuggability, bincode for size — tracked as a minor decision, not surfaced).

```
// helixvpn-ipc/src/lib.rs - SHARED crate compiled into BOTH the
// Rust service and (via cbindgen
// or a hand-mirrored Dart type) the Dart client, so the wire
// types can never drift (§12 UT).
#[derive(Serialize, Deserialize)]
pub enum PipeFrame {
    Req { id: u64, cmd: Command },           // UI → service
        (request/reply correlated by id)
    Resp { id: u64, result: CmdResult },     // service → UI (reply
        to a Req)
```

```

        Event(PipeEvent),                                // service → UI
        (unsolicited push: status + lifecycle)
    }

#[derive(Serialize, Deserialize)]
pub enum Command {                                     // mirrors
    TunnelPlatform [03-client §4] + the FFI surface [03-client
        §3]
    StartTunnel(TunnelConfig),                          // §4.5
    StopTunnel,
    SetShields(Shields),                                // logic-only,
        forwarded to core.set_shields
    SetExit { id: String, multi_hop_chain: Option<Vec<String>> },
    Exits,                                                // → Vec<ExitOption>
    Advertise(Vec<String>),                              // connector flavor
    Hello { protocol_version: u32, client_token: [u8; 32] }, //
        §4.3 handshake
    Ping,                                                // liveness keepalive
        (§4.6)
}

#[derive(Serialize, Deserialize)]
pub enum CmdResult {
    Ok,
    Exits(Vec<ExitOption>),
    Advertised(AdvertiseResult),
    HelloOk { protocol_version: u32, service_version: String },
    Err(PipeError),                                     // §11 error taxonomy
        (never carries a secret, W-I6)
}

#[derive(Serialize, Deserialize)]
pub enum PipeEvent {
    Status(TunnelStatus),                                // the CANONICAL core
        enum (§3.3) [01-orch §4.1]
    Platform(PlatformTunnelEvent),                       // up/down/
        permissionDenied/revoked/error [03-client §4]
}

```

Why both Status and Platform events. TunnelStatus carries the *core's* truth (it is connecting/connected/reconnecting); PlatformTunnelEvent carries the *shim's* lifecycle truth (adapter up/down, the service was revoked/stopped out-of-band, a privileged step errored). The TunnelPlatform contract [03-client §4] is realised by mapping these two event streams into its events() stream (§5.2). They are distinct because a permissionDenied/error can occur *before* the core ever emits a TunnelStatus (e.g. WFP install failed).

4.3 Pipe security descriptor + client authentication (W-I4)

Two layers guard the pipe:

1. **DACL on the pipe object.** CreateNamedPipeW is given a SECURITY_ATTRIBUTES whose security descriptor (SDDL) grants **connect/**

read/write only to the interactive desktop user(s) and full control to SYSTEM + Administrators, and **denies NETWORK** (no remote pipe access — pipes are reachable over SMB by default; this **MUST** be blocked):

```
D: (D;;;FA;;;NU) (A;;;0x12019b;;;IU) (A;;;FA;;;SY) (A;;;FA;;;BA)
; NU = NETWORK USERS → DENY ; IU = INTERACTIVE USERS →
connect+rw ; SY/BA → full
```

UNVERIFIED: exact access mask 0x12019b (= GENERIC_READ | GENERIC_WRITE | SYNCHRONIZE minus FILE_CREATE_PIPE_INSTANCE) is a recommendation to pin against winnt.h, calibrated so a client can read/write but **cannot create a second server instance**.

2. **Token check on connect.** On ConnectNamedPipe, the service calls GetNamedPipeClientProcessId → opens the client process token → asserts the client runs in the **same interactive session** (WTSGetActiveConsoleSessionId) and is not elevated beyond what is expected, then requires the first frame to be a Command::Hello carrying a client_token minted at install/launch time. A connection that does not Hello within HELLO_TIMEOUT (recommend 2 s, UNVERIFIED) is dropped (§11). This defeats a malicious local process racing to the pipe before the real UI.

4.4 The event pump (core StreamSink → pipe)

status_stream(sink) (§3.1) is handed a StreamSink<TunnelStatus> whose implementation, in the service, **serialises each TunnelStatus into a PipeFrame::Event(PipeEvent::Status(...)) and writes it to every connected pipe instance**. The service also injects PipeEvent::Platform(...) for adapter/WFP/service-lifecycle transitions (§7). Backpressure: the pipe write uses a bounded per-client queue (recommend 64 frames, mirroring the STATUS_CHANNEL_CAP of [01-orch §4.2]); a wedged client that overflows is **disconnected** (not blocked) so a stuck UI never stalls the core — the UI reconnects and re-reads latest (§5.3), exactly the Lagged → re-read latest discipline of [01-orch §4.6].

4.5 TunnelConfig on the wire (consistent with [03-client §4])

```
#[derive(Serialize, Deserialize, Clone)]
pub struct TunnelConfig {                                // mirrors Dart
    TunnelConfig [03-client §4]
    pub overlay_ip: String,                               // "100.64.0.7/32" |
        "fd7a:.../128"
    pub routes: Vec<String>,                             // AllowedIPs →
        wintun route table (iphlpapi)
    pub dns_servers: Vec<String>,                         // tunnel DNS (§8.3
        leak guard)
    pub split_exclude_apps: Vec<String>,                 // per-app bypass →
        WFP app-id filters (§8.5)
    pub split_exclude_routes:
        Vec<String>, // per-route bypass → route-table
        exclusions (§8.5)
    pub mtu: u16,                                         // default 1280 over
        MASQUE / 1420 plain WG
```

```

pub session_or_map_token: String, // →
  ClientConfig.map_path_or_session [03-client §3.1] (W-I6:
    never logged)
}

```

4.6 Pipe-level liveness & reconnect

- **Keepalive.** The UI sends `Command::Ping` every `PING_INTERVAL` (recommend 5 s, UNVERIFIED); the service replies `CmdResult::Ok`. A missed reply past `PING_TIMEOUT` (recommend $3 \times$ interval) $\Rightarrow$ the UI treats the pipe as dead, tears it, and reconnects.
- **Service-restart resilience.** If the service restarts (crash $\rightarrow$ SCM recovery, §6.4) the pipe handle breaks; the UI's read returns `ERROR_BROKEN_PIPE`. The UI reconnects with backoff and re-issues `Hello`; the service replays the **current** `TunnelStatus` as the first event so the UI re-syncs (W-I5). The *tunnel itself is unaffected* by a UI/pipe drop — the core keeps forwarding (the pipe is control-only, never in the packet path; the Windows analogue of the fail-static I3 [01-orch §0.3]).

5. The TunnelPlatform contract implementation (Dart side)

On Windows, `WindowsTunnelPlatform` is the `TunnelPlatform` [03-client §4] impl whose body is "marshal the verb to the pipe, fold the pipe's event stream into `events()`" [03-client §5.3].

5.1 The Dart implementation

```

// helix_core_ffi/lib/src/windows/windows_tunnel_platform.dart
class WindowsTunnelPlatform implements TunnelPlatform { //
  contract: [03-client §4]
  final _PipeClient _pipe; //
    dart:ffi → kernel32 (§5.1.1)
  final _events =
    StreamController<PlatformTunnelEvent>.broadcast();

  @override
  Future<void> startTunnel(TunnelConfig cfg) async {
    await _pipe.ensureConnected(); //
      CreateFileW + Hello (§4.3)
    final r = await _pipe.request(Command.startTunnel(cfg)); //
      01: idempotent, permission-aware
    if (r is CmdErr) throw TunnelException(r.error); //
      never a half-open adapter (01)
  }

  @override
  Future<void> stopTunnel() => _pipe.request(const
    Command.stopTunnel()); // 04: quiescent restore

```

```

@override
Stream<PlatformTunnelEvent> events() => _events.stream;    //
    03: surfaces revoked/error

// wiring: pipe events → TunnelPlatform.events() + the
    helix_core status stream (§5.2)
void _onPipeEvent(PipeEvent e) => switch (e) {
    StatusEvent(:final status) =>
        _statusController.add(status),    // →
        helix_core.statusStream()
    PlatformEvt(:final platform) =>
        _events.add(platform),    // →
        TunnelPlatform.events()
};
}

// §5.1.1 – the pipe client is a thin dart:ffi binding to kernel32
    (no third-party plugin needed):
//   CreateFileW(r'\\.\pipe\helixvpn\tunnel', GENERIC_READ|
        GENERIC_WRITE, 0, null, OPEN_EXISTING,
        FILE_FLAG_OVERLAPPED, null)
//   ReadFile / WriteFile with the 4-byte length-prefix framing
    (§4.2), on a dedicated isolate so
//   the overlapped reads never block the UI isolate. UNVERIFIED:
    overlapped-IO ergonomics from Dart
//   vs a blocking read on a background isolate – pick by the FA/
    IT smoke (§12); a small native
//   helper DLL is the fallback if dart:ffi overlapped-IO proves
    awkward.

```

5.2 Mapping the two pipe event streams to the doc-03 surface

WindowsTunnelPlatform feeds **two** Dart sinks from the **one** pipe event stream, and the Dart helix_core_ffi wrapper composes them into the **richer doc-03 TunnelStatus** (Disconnected/path/Danger) [03-client §3.1]:

Pipe event	Drives	Doc-03 UI surface
Status(Connecting/Handshaking/Reconnecting)	helix_core.statusStream()	amber stateConnecting [03-client §7.1]
Status(Connected{transport,rtt})	helix_core.statusStream()	green stateConnected; StatusChip shows transport·rtt
Status(Down{reason})	helix_core.statusStream()	mapped to Disconnect (reason "stopped") or Danger/Down palette (any other reason) [03-client §7.1]
Platform(revoked)	TunnelPlatform.events()	Danger{kind:"revoked" overlay — admin/OS kill the tunnel (O3)
Platform(permissionDenied)	TunnelPlatform.events()	

Pipe event	Drives	Doc-03 UI surface
		a permission/elevation prompt, never silent failure (O1)
Platform(error, detail)	TunnelPlatform.events()	honest error surface with detail (§11.4.6 no guessing)

This is the W-I5 + CI2 honesty guarantee made concrete: the green palette is painted **only** on a core Connected, never on UI intent [03-client §8.4].

5.3 Contract obligations on Windows (the O1–O5 of [03-client §4.1])

O#	Windows realisation
O1 idempotent + permission-aware startTunnel	a second StartTunnel while connected is a no-op 0k; if the service is not installed / not elevated, the UI emits permissionDenied (offering to (re)install/repair the service via an elevated installer), never a half-open adapter — the service rolls back the adapter + WFP on any start failure (§7.4)
O2 shim hands the core packets, never crypto	the service pumps the wintun ring into helix-core; WG + transport live only in the core (W-I2)
O3 report revoked out-of-band	device.revoked (doc 02) → core Down{auth-failed} [01-orch §5.4] → Platform(revoked); an admin/MDM sc stop → Platform(down); SCM SERVICE_CONTROL_STOP → graceful teardown event
O4 quiescent restore on stop	on StopTunnel the service removes routes, reverts DNS, deletes dynamic split-tunnel WFP filters, and (user stop only) removes the persistent kill-switch filters, then ends the wintun session — on every exit path (Drop/finally), §8.4
O5 the only untyped seam, covered on real hardware	the pipe + adapter + WFP are exercised by a real-Windows-VM smoke (§12 IT/SC), never a mock-claiming-PASS (§11.4.27)

6. Windows service lifecycle (SCM state machine)

6.1 Service identity & install

Property	Value
Service name	HelixTunnelSvc Helix VPN Tunnel Service

Property	Value
Display name	
Binary	helix_tunnel_svc.exe (Authenticode-signed, §9)
Account	LocalSystem (or hardened virtual account, §1.1 / D-WIN-3)
Start type	SERVICE_AUTO_START (so a tunnel can auto-reconnect at boot) or DEMAND_START (privacy default — only runs when the UI asks) — D-WIN-4 (§13)
Error control	SERVICE_ERROR_NORMAL
Recovery	restart on failure with backoff (§6.4)

Install is performed by the **signed installer** (MSI/MSIX, §9) via the Service Control Manager (CreateServiceW / ChangeServiceConfig2W for recovery actions), **never** by the unprivileged UI. The installer also drops wintun.dll (+ optional wireguard.sys) into the service directory and stages the pipe client_token (§4.3) in a per-machine location the UI can read.

Service authoring language — D-WIN-2 (§13). Recommended: a **Rust** service binary (windows-service crate for the SCM ServiceMain/ ServiceCtrlHandler plumbing) so helix-core is hosted **in-process with zero intra-process FFI** and no GC on the datapath (W-I2, byte-for-byte core reuse). The doc-03 "C#/Rust" note [03-client §5.3] is honoured as the alternative: a **.NET Worker Service** hosting the core as a native DLL via P/Invoke — viable but adds a CLR + a marshalling seam. Both are surfaced; Rust is the recommendation.

6.2 SCM state machine

stateDiagram-v2

```

[*] --> Stopped : installed (CreateServiceW)
Stopped --> StartPending : SCM Start (UI requests via
installer-staged trigger / sc start)
StartPending --> Running : ServiceMain ready – pipe server up,
core idle, kill-switch ARMED per config (§8.2)
StartPending --> Stopped : init failed (driver missing / WFP
denied) → SetServiceStatus(STOPPED, win32_exit)
Running --> StopPending : SCM Stop (SERVICE_CONTROL_STOP) OR
last UI graceful StopTunnel+idle-timeout
StopPending --> Stopped : teardown done (§8.4) – adapter
ended, DNS reverted, kill-switch per §8.6
Running --> Running : SERVICE_CONTROL_INTERROGATE /
SESSIONCHANGE (logon/logoff) – re-evaluate pipe ACL
Running --> [*] : SERVICE_CONTROL_SHUTDOWN (machine shutdown)
– fast quiescent stop

```

6.3 ServiceMain skeleton

```

// helix_tunnel_svc/src/main.rs (Rust, windows-service crate)
fn service_main(_args: Vec<OsString>) {

```

```

let status_handle =
    service_control_handler::register("HelixTunnelSvc",
        ctrl_handler)?;
set_status(&status_handle, StartPending);
let rt =
    tokio::runtime::Builder::new_multi_thread().enable_all().build()?;
rt.block_on(async {
    let pipe = PipeServer::create(PIPE_NAME,
        security_descriptor())?; // §4.1/§4.3
    let svc =
        ServiceState::new(); //
        owns Option<Orchestrator>
    set_status(&status_handle, Running);
    svc.serve(pipe,
        shutdown_rx).await; //
        accept clients, dispatch Commands
});
set_status(&status_handle, Stopped);
}

extern "system" fn ctrl_handler(control: ServiceControl) ->
    ServiceControlHandlerResult {
    match control {
        ServiceControl::Stop | ServiceControl::Shutdown => {
            signal_shutdown(); Handled }
        ServiceControl::Interrogate => Handled,
        ServiceControl::SessionChange(_) => { reeval_pipe_acl();
            Handled } // logon/logoff
        _ => NotImplemented,
    }
}

```

6.4 Crash recovery & fail-closed coupling (W-I3)

SCM recovery actions (SC_ACTION_RESTART) restart the service after a crash with backoff (recommend: 1st failure +1 s, 2nd +5 s, subsequent +30 s, reset window 24 h — UNVERIFIED, tune). **Critically:** because kill-switch WFP filters are **persistent** (W-I3, §8.4), a crash leaves egress blocked until the restarted service either re-establishes the tunnel (→ open via tunnel) or a user graceful-stop removes them — there is **no leak window across a crash**. On restart the service re-reads its last desired state and re-dials (the core's own reconnection backoff [01-orch §7] then governs).

7. Tunnel lifecycle on Windows (end-to-end sequence)

```

sequenceDiagram
    autonumber
    participant UI as Flutter UI (user)
    participant PIPE as named pipe
    participant SVC as HelixTunnelSvc (SYSTEM)

```


participant CORE as helix-core (in-service)
participant WT as wintun adapter
participant WFP as WFP engine
participant GW as gateway :443/:51820

```
UI->>PIPE: Hello{proto, client_token}           %% §4.3
handshake
PIPE->>SVC: verify session+token → HelloOk
UI->>PIPE: StartTunnel(TunnelConfig)             %% §4.5
PIPE->>SVC: Command::StartTunnel
SVC->>WFP: install PERSISTENT kill-switch (closed: allow gw
endpoints only)  %% §8.2 (W-I3)
SVC->>WT: WintunCreateAdapter + StartSession + set IP/routes/
DNS (iphlpapi)
SVC->>CORE: start(ClientConfig{token,...}) +
attach_tun_pump(reader,writer)  %% §3.1/§3.2
SVC-->>PIPE: Event Platform(up)
SVC-->>PIPE: Event Status(Connecting)            %% core enum,
verbatim (W-I5)
CORE->>GW: dial ladder rung → WG handshake [01-trans §6]
SVC-->>PIPE: Event Status(Handshaking)
GW-->>CORE: WG handshake reply
SVC->>WFP: OPEN kill-switch (egress via tunnel) + apply DNS
leak block  %% §8.4
SVC-->>PIPE: Event Status(Connected{transport:"masque-h3",
rtt_ms:23})
Note over CORE,WT: loops pump wintun ring ≠ WG ≠ transport
[01-orch §3]
UI->>PIPE: SetShields(...) / SetExit(...)         %% logic-
only, no adapter touch (CI1)
PIPE->>SVC: → core.set_shields / set_exit
UI->>PIPE: StopTunnel
PIPE->>SVC: Command::StopTunnel
SVC->>CORE: stop() + detach_tun()
SVC->>WT: WintunEndSession + WintunCloseAdapter; revert
routes/DNS
SVC->>WFP: delete DYNAMIC split filters; remove PERSISTENT
kill-switch (user stop only, §8.6)
SVC-->>PIPE: Event Status(Down{reason:"stopped"}) +
Platform(down)
```

The seam mirrors the cross-platform rule [03-client §3.3]: **lifecycle commands flow UI → pipe → service → core, status events flow core → service → pipe → UI**, and setShields/setExit/exits are pure logic that never touch the adapter or WFP (CI1).

7.1 Start ordering (the no-leak ordering)

1. Install **persistent closed** kill-switch (egress blocked except gateway endpoints) — *before* the adapter exists, so there is no pre-connect leak window (§8.2, mirrors [01-orch §8.2]).
2. Create + configure the wintun adapter (IP/routes/DNS).

3. `core.start(...)` + `attach_tun_pump(...)`; the orchestrator drives the ladder.
4. On core Connected: **open** the kill-switch to route via the tunnel + apply the DNS-leak block (§8.4). The UI sees green only here.

7.2 Stop ordering (the quiescent restore, O4)

stop reverses: `core.stop()` → end wintun session → revert routes/DNS → delete dynamic split filters → (user stop only) remove persistent kill-switch → emit `Down{stopped}`. On a **crash** or **non-user terminal** (ladder-exhausted, auth-failed, host-fatal [01-orch §4.4/§8.4]) the persistent kill-switch **stays** (W-I3 / O-I11).

7.3 Idempotency & rollback

Every privileged step is paired with a rollback registered in reverse order (a Rust Drop-guard stack). If step 3 fails (e.g. dial never succeeds within the ladder budget), the guard stack runs: detach core → end session → close adapter → keep kill-switch closed (no leak) → emit `Platform(error, detail)` + `Status(Down{reason})`. There is never a half-configured adapter left behind (O1).

8. WFP — kill-switch + split-tunnel

The Windows realisation of the core's kill-switch + DNS coupling [01-orch §8] and the TunnelConfig split fields (§4.5) is **WFP (Windows Filtering Platform)** — filters installed by the service against the FWPM filter engine, driven by the core's OrchState transitions [01-orch §8.4] (never hand-edited).

8.1 WFP object model used

WFP object	Use
Engine handle (<code>FwpmEngineOpen0</code>)	one per service; opened at start, closed at stop
Persistent sub-layer (<code>FWPM_SUBLAYER</code> + <code>FWPM_FILTER_FLAG_PERSISTENT</code>)	the kill-switch filters — survive a service crash / reboot (W-I3, fail-closed)
Dynamic sub-layer (<code>FWPM_SESSION_FLAG_DYNAMIC</code>)	the split-tunnel exclusion filters — auto-removed when the engine handle closes with the tunnel
Layers	<code>FWPM_LAYER_ALE_AUTH_CONNECT_V4/_V6</code> (per-connection allow/block, incl. app-id conditions) and <code>FWPM_LAYER_OUTBOUND_IPPACKET_V4/_V6</code> (catch-all egress)
Conditions	<code>FWPM_CONDITION_IP_REMOTE_ADDRESS</code> (gateway allow), <code>FWPM_CONDITION_IP_LOCAL_INTERFACE</code> (the wintun LUID),

WFP object	Use
	FWPM_CONDITION_ALE_APP_ID (per-app split), FWPM_CONDITION_IP_REMOTE_PORT (:53 DNS leak guard)

UNVERIFIED: exact GUIDs / struct layouts pinned against the Microsoft fwpmu.h / fwpmypes.h headers; the model above is the documented WFP usage pattern (the same one WireGuard's own Windows kill-switch uses).

8.2 Kill-switch filters (persistent, fail-closed)

On Idle → Connecting (Strict mode [01-orch §8.2]) the service installs, in the persistent sub-layer, a **default-block** egress filter (high weight) plus narrow **permit** filters:

Permit filter	Why
remote-address ∈ allow_gateway_endpoints [01-orch §8.2]	the transport must reach :443/:51820
local-interface = wintun LUID	tunnelled traffic is always allowed
DHCP / ND (bootstrap)	link bring-up
optional: RFC1918 LAN if allow_lan	printers etc. [01-orch §8.2]

Everything else egress is **blocked**. A WFP-install failure at start ⇒ Down{host-fatal} [01-orch §4.4] — never proceed on an unverified kill-switch (§11.4.6). Because these filters are persistent, a service crash leaves them installed (W-I3).

8.3 DNS-leak guard

On Handshaking → Connected [01-orch §8.4] the service (if block_plaintext_53) installs a filter dropping any :53 egress (UDP+TCP) **not** destined for the tunnel DNS over the wintun interface, and applies the tunnel DNS via SetInterfaceDnsSettings on the adapter LUID. The block **stays** across Connected → Reconnecting/Down (fail-closed) until reconnect re-applies tunnel DNS or a user stop reverts to system resolvers (§8.6) — exactly [01-orch §8.3].

8.4 Coupling table (Windows realisation of [01-orch §8.4])

Core transition [01-orch §5.2]	Kill-switch (persistent)	DNS / split (dynamic)
Idle → Connecting (Strict)	install block + gateway permits	leave system DNS
Handshaking → Connected	add tunnel permit (egress via wintun)	apply tunnel DNS + :53 block + install split-tunnel excludes
Connected → Reconnecting		keep :53 block; keep split excludes

Core transition [01-orch §5.2]	Kill-switch (persistent)	DNS / split (dynamic)
	revoke tunnel permit (block holds)	
* → Down{reason≠stopped}	stays blocked	keep block
ShuttingDown → Down{stopped} (user)	remove all kill-switch filters	revert system DNS; delete split excludes
crash / panic	persistent filters remain	dynamic excludes auto-removed (engine handle closed) — kill-switch still blocks

The crash asymmetry is deliberate (W-I3). Kill-switch = persistent → *survives* crash (fail-closed, no leak). Split-tunnel excludes = dynamic → *vanish* on crash (so a crash can only make traffic *more* tunnelled / blocked, never *less* protected). This is the WFP encoding of O-I11.

8.5 Split-tunnel (per-app + per-route)

- **Per-route** (split_exclude_routes): the excluded CIDRs are **removed from the wintun route table** (they egress on the physical adapter) AND a dynamic WFP *permit* filter lets them bypass the kill-switch. Simplest, app-agnostic.
- **Per-app** (split_exclude_apps): a dynamic FWPM_CONDITION_ALE_APP_ID filter at the ALE_AUTH_CONNECT layer permits the named executable's connections to bypass the tunnel. app_id is the device-path form of the binary (FwpmGetAppIdFromFileName0). UNVERIFIED: full per-app split fidelity (vs route-based) is a Phase-2 item — Phase-1 ships per-route split, per-app behind a capability flag (Capability.splitTunnel [03-client §6]).

8.6 The only open path

The persistent kill-switch is removed on **exactly one** path — a user-initiated graceful stop (StopTunnel → ShuttingDown → Down{stopped}) — identical to the core rule [01-orch §8.6]. Every other terminal leaves it closed.

9. Code signing & driver trust (§11.4.133)

§11.4.133 (target-system safety) + the doc-03 signing note [03-client §11] require **no unsigned privileged code**. Three trust tiers:

Artifact	Signing requirement
wintun.dll / wireguard.sys	Microsoft/WireGuard-LLC signed drivers, shipped as-is — we do not re-sign third-party drivers. If a custom driver were ever needed it would require EV + Microsoft attestation signing (Partner

Artifact	Signing requirement
	Center hardware dashboard) — out of scope for MVP (we use the upstream signed wintun).
helix_tunnel_svc.exe + DLLs	Authenticode signed with the project's EV (or OV) code-signing cert , SHA-256, RFC-3161 timestamped so signatures outlive cert expiry.
Installer (MSI/MSIX)	Authenticode signed (same cert); MSIX additionally for Store/sideload trust.

9.1 Signing pipeline (local, §11.4.156 no server CI)

```
# local pre-tag step (§11.4.40 sweep) – operator runs with the EV
token/HSM present
signtool sign /fd SHA256 /tr http://timestamp.acme/rfc3161 /td
SHA256 /a helix_tunnel_svc.exe
signtool sign /fd SHA256 /tr http://timestamp.acme/rfc3161 /td
SHA256 /a HelixVPN-Setup.msi
signtool verify /pa /v helix_tunnel_svc.exe          # MUST pass
before the artifact ships
```

UNVERIFIED: the cert is **operator-supplied** (§11.4.10 — the private key lives on an EV token/HSM, never in the repo, never in .env tracked files). SmartScreen reputation accrues over EV-signed releases. The signing step is a **release-gate blocker**: an unsigned helix_tunnel_svc.exe cannot ship (§11.4.133). The verification output (signtool verify /pa) is the captured evidence (§12 SEC).

9.2 What signing buys (honest boundary, §11.4.6)

Signing proves **provenance + integrity** (the binary is ours, untampered) and is required for the driver to load + for SmartScreen/UAC trust. It does **not** prove the code is *correct* or *leak-free* — that is the §12 test matrix's job. Do not conflate "signed" with "safe".

10. Memory / resource budgets

Windows has **no iOS-NE-class ~15 MB ceiling** [03-client §0.1 CI4]; the service is a normal desktop process. Budgets are nonetheless bounded (host-safety adjacency, §12.6):

Budget	Target (recommendation, UNVERIFIED — calibrate in BENCH §12)	How
Service idle RSS	< ~30–40 MB	Rust service (no CLR/GC); reused encrypt/decrypt scratch buffers per loop (no per-packet alloc, [01-orch §3.1]); LTO + strip release build

Budget	Target (recommendation, UNVERIFIED — calibrate in BENCH §12)	How
wintun ring capacity	4 MiB (0x400000, power-of-two within 0x20000..=0x4000000)	bounded send/recv ring; back-pressure rather than unbounded growth
Pipe per-client queue	64 frames (mirrors STATUS_CHANNEL_CAP [01–orch §4.2])	overflow ⇒ disconnect the wedged client, never block the core (§4.4)
Status broadcast cap	64 (the core's [01–orch §4.2])	unchanged from the cross-platform core
UI (unprivileged) idle	single-digit → low-tens MB	Flutter AOT, no core linked in the UI process (the core is in the service)
CPU per Gbps	per the transport BENCH [01–trans §11]	core is identical to every platform; the shim adds only the wintun copy + pipe marshalling of <i>control</i> frames (not packets)

The pipe never carries packets. Data packets flow wintun ⇔ core ⇔ network entirely inside the service; only *control* (commands + status events) crosses the pipe. The pipe is therefore low-bandwidth and never a throughput bottleneck (W-C2).

11. Error taxonomy & edge cases

```
// helixvpn-ipc / helix-core/src/platform/windows – surfaced as
// PipeError on the wire + PlatformTunnelEvent
#[derive(Serialize, Deserialize, Debug)]
pub enum PipeError {
    ServiceNotInstalled,           // UI: offer elevated (re)install
    → PlatformTunnelEvent.permissionDenied
    NotElevatedToInstall,         // installer needs admin
    AdapterCreateFailed(String), // wintun WintunCreateAdapter
    failed (driver missing / name clash)
    WfpInstallFailed(String),     // kill-switch could not be
    installed → Down{host-fatal} (never proceed)
    PipeAuthRejected,            // Hello failed / wrong session /
    token mismatch (§4.3)
    ProtocolVersionMismatch { service: u32, client: u32 },
    Busy,                         // too many pipe instances / a
    start already in flight
    CoreError(String),           // forwarded anyhow::Error from
    the FFI surface (never a secret, W-I6)
}
```

11.1 Edge-case matrix

#	Edge case	Required behaviour
E1	Service not installed / stopped when UI calls <code>startTunnel</code>	<code>permissionDenied</code> + offer elevated installer/repair; never silent failure (O1)
E2	UI process killed mid-tunnel	pipe breaks; tunnel keeps running (control-only pipe, §4.6); UI reconnects + re-syncs <code>TunnelStatus</code>
E3	Service crashes mid-tunnel	SCM restarts it (§6.4); persistent kill-switch holds → no leak (W-I3); core re-dials on restart
E4	Two UIs (or a malicious local proc) race the pipe	DACL + Hello token gate (§4.3); excess instances → <code>Busy</code> ; <code>NETWORK</code> denied (W-I4)
E5	wintun driver absent / wrong version	<code>AdapterCreateFailed</code> → installer repair flow; never a half-open adapter
E6	WFP filter install denied (policy/AV)	<code>WfpInstallFailed</code> → <code>Down{host-fatal}</code> ; do not run with a believed-but-unverified kill-switch (§11.4.6)
E7	Machine sleep/resume; NIC change (Wi-Fi→Ethernet)	the core's <code>RoamDetected</code> fast-path [01-orch §7.5] re-dials; the wintun adapter persists across the roam (WG roams on top)
E8	User logoff while tunnel up (auto-start mode)	service keeps running headless; pipe ACL re-evaluated on <code>SESSIONCHANGE</code> (§6.3); UI reconnects on next logon
E9	Machine shutdown	<code>SERVICE_CONTROL_SHUTDOWN</code> → fast quiescent stop; persistent kill-switch is fine to leave (next boot re-arms)
E10	Oversize datagram on MTU change (transport swap)	core lowers inner MTU / re-fragments ([01-trans §4.8] rule 3); shim re-applies adapter MTU via <code>iphlpapi</code>

11.2 No-guessing classification (§11.4.6)

A pipe/adaptor/WFP failure is **never** reported as "transient" without captured evidence; each `PipeError` variant maps to a concrete observed signal (a Win32 error code, a WFP `HRESULT`, a wintun return). Ambiguous "the UI lost the pipe" is `ERROR_BROKEN_PIPE` → reconnect, not a guess about *why* — the core's `TunnelStatus` remains the truth about the *tunnel* (W-I5).

12. Test points — §11.4.169 closed test-type vocabulary

Every Windows-shim workable item declares its required test types from the §11.4.169 closed set; the only permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason, never a silent gap. Four-layer enforcement per §11.4.4(b)

applies to every closure. Captured evidence per §11.4.5/.69/.107; the shim is the only untyped seam and is covered on a **real Windows VM / machine**, never a mock-claiming-PASS (O5, §11.4.27).

Code (§11.4.169)	Concrete Windows-shim test point	Evidence
UT	PipeFrame/Command/TunnelStatus serde round-trip is identity; the Dart mirror types match the Rust helixvpn-ipc types byte-for-byte (drift FAILs build); PipeError mapping; SDDL string builds the intended DACL	cargo test + Dart flutter test output; mirror-diff
IT	on a real Windows VM (booted via the containers submodule §11.4.76 where virtualisation allows, else a pinned VM runner): install the service, create a wintun adapter, open the pipe, drive Hello → StartTunnel → Connected against a real edge	adapter-up log + pipe transcript + per-carrier round-trip
E2E	UI → pipe → service → core → wintun → gateway: curl to a LAN host succeeds over plain-udp and over masque-h3 (with plain WG nft-blocked at the edge)	window-scoped MP4 (§11.4.159) of the connect journey + curl capture
FA	one-shot scripted install→connect→shields→stop→uninstall, N=3 identical runs producing the same ordered TunnelStatus trace (deterministic §11.4.50)	3× identical status-trace artifacts
CH	a Challenge (challenges submodule §11.4.27(B)) scores the captured E2E evidence (real connect, not config)	Challenge result.json
HQA	HelixQA autonomous session drives install→connect→roam→stop across carriers	HelixQA session evidence
LOAD	rapid connect/disconnect churn (100×) + many concurrent pipe Hello attempts: no adapter leak, no handle leak, bounded RSS	handle-count + RSS CSV
SEC	(a) signtool verify /pa passes on the service + installer (§9); (b) pipe DACL denies a NETWORK/non-interactive caller (W-I4); (c) plant the session token + a SecretBytes → grep proves it appears in no log/tracked file (§11.4.10, W-I6); (d) kill-switch: pull the carrier, assert zero egress leaks via packet capture	signtool output; rejected-connect log; grep-empty proof; leak-capture
SC	crash the service mid-tunnel → assert persistent kill-switch still blocks (W-I3) + SCM restart recovers < 3 s [01-orch §7.1]; NIC flap (Wi-Fi→Ethernet) → roam recovery; kill the UI → tunnel survives (E2)	recovery trace + leak-during-crash capture (MUST be empty)
CONC	concurrent pipe clients (allowed N) + atomic transport swap under load → zero dropped control frames; loop A send / loop B recv during a wintun-ring contention burst	concurrency harness log

Code (§11.4.169)	Concrete Windows-shim test point	Evidence
RACE	cargo test --features loom on the pipe per-client queue + the service ServiceState shared cell; pipe-connect race vs Hello token gate	loom report
MEM	service RSS bound under churn (§10); no per-datagram heap growth on the wintun pump; no handle leak across 100 connect cycles	/ proc-equivalent (PerfMon) RSS + handle CSV
BENCH	per-carrier throughput / CPU-per-Gbps through the wintun path; wintun vs (optional) wireguard-nt plain-udp fast-path delta (informs D-WIN-1)	throughput/CPU/ p99 CSV

12.1 Anti-bluff rule (§11.4.5/.69/.107)

A green UT serde test does **not** prove the tunnel works — the **IT/E2E/SC layers on a real Windows machine are the authoritative evidence**: the window-scoped MP4 of the real connect journey (StatusChip reading masque-h3 · 23ms, OCR/vision-validated §11.4.159) + the **empty** leak capture during a forced crash (W-I3) + the signtool verify pass. The kill-switch claim in particular is proven by a packet capture showing zero egress while the carrier is down, never asserted (§11.4.6/.123).

13. Open decisions surfaced by this document

Per §11.4.6/§11.4.66 — options + recommendation, never silently resolved. These are *Windows-shim* decisions; program-level D1–D8 live in doc 00, client-stack D-CLIENT-1..5 in [03-client §14].

#	Decision	Option A	Option B	Recommendation
D-WIN-1	Primary tunnel adapter	wintun + core-owned WG/transport (every obfuscation carrier works, W-I2)	wireguard-nt kernel WG (plain-udp only, in-kernel speed)	A for the datapath ; wireguard-nt as an <i>optional</i> Phase-2 plain-udp kernel fast path, gated on the BENCH delta. Ship wintun-only in Phase 1.
D-WIN-2	Service language	Rust (windows-service, core in-process, no GC/marshalling)	C# .NET Worker Service (core via P/Invoke)	A — byte-for-byte core reuse (I4), smaller, no CLR; honours the doc-03 "C#/Rust" note as the alternative.

#	Decision	Option A	Option B	Recommendation
D-WIN-3	Service account	LocalSystem (simplest, certainly works)	dedicated virtual service account NT SERVICE\HelixTunnelSvc with minimal privileges	B if a SEC/IT measurement (§12) confirms wintun + WFP work under the reduced token; else A. Surface the measurement, do not assume.
D-WIN-4	Service start type	DEMAND_START (privacy: runs only when asked)	AUTO_START (reconnect-at-boot convenience)	A default (least-surprise/privacy); offer AUTO_START as an explicit user opt-in for always-on.
D-WIN-5	Pipe body encoding	MessagePack (rmp-serde, compact, self-describing)	JSON (debuggable) / bincode (smallest)	A — compact + cross-language; the helixvpn-ipc crate freezes the schema so Dart + Rust cannot drift (§12 UT).

Sources verified

- `final/03-client-core-and-ui.md` [03-client] — §0.1 (client invariants CII–CI6), §3 (FFI surface, TunnelStatus/Shields/ExitOption types, the iOS pump pattern this doc mirrors for wintun), §4 (TunnelPlatform contract + TunnelConfig + PlatformTunnelEvent + obligations O1–O5), §5.1 (iOS packetFlow $\approx$ core pump — the model for the wintun pump), §5.3 (Windows row: wireguard-nt/wintun + privileged service + named-pipe IPC + WFP + Authenticode), §6 (capabilities incl. splitTunnel), §7.1 (state palette), §8.4 (intent-vs-actual honesty), §11 (signing), §12 (testing + anti-bluff).
- `final/v02-data-plane/orchestrator-and-state.md` [01-orch] — §0.3 (invariants incl. O-I11 fail-closed), §2.1 (TunDevice role wintun fills), §3 (the three loops — unchanged on Windows), §4.1 (the canonical five-variant TunnelStatus the pipe carries), §4.2 (channel cap), §4.4 (Down.reason vocabulary), §4.6 (receiver Lagged $\rightarrow$ re-read latest), §5.2 (state diagram), §7 (reconnection backoff + roam fast-path), §8 (kill-switch + DNS coupling — the WFP realisation in §8 here).
- `final/v02-data-plane/transport-trait.md` [01-trans] — §1 (invariant I4: one core, three consumers — why wintun not wireguard-nt for obfuscated carriers), §3.2 (SecretBytes no-leak), §4.8 (MTU/Oversize), §6 (the ladder the in-service core runs), §11 (perf budget).
- `04_VPN_CLD/HelixVPN-helix-ui-Flutter.md` [04_UI] — §5/§6 (per-platform shim matrix: Windows = wireguard-nt/wintun + privileged service + named-pipe IPC, C#/Rust; signing note).

- 04_VPN_CLD/HelixVPN-Architecture-Refined.md [04_ARCH] — §5.3 (shim matrix), §5.6 (kill switch + DNS-leak driven by core state), §5.7 (desktop posture).
- [SYN] cross-document synthesis — §5 (client architecture: Windows wireguard-nt/wintun + privileged service + named-pipe IPC), §9 (constitution bindings).

Constitution: §11.4.44 (revision header), §11.4.6/§11.4.66 (decisions = options + recommendation; no guessing), §11.4.10 (no secret leak — SecretBytes/token never logged, W-I6), §11.4.27 (no fakes beyond unit tests — the shim covered on real hardware), §11.4.5/§11.4.69/§11.4.107/§11.4.159 (captured / window-scoped evidence), §11.4.133 (signed privileged code / target safety — §9), §11.4.156 (no active CI — local signing/test ritual), §11.4.169 (comprehensive test-type coverage — §12).

End of nano-detail specification — Windows shim (wireguard-nt + service), Volume 4 (Clients). Pairs with [03-client §5.3] (the overview this deepens), [01-orch §4.1/§8] (the TunnelStatus + kill-switch contracts this realises on WFP), and the sibling per-platform shim docs. Surfaced decisions: D-WIN-1 (adapter), D-WIN-2 (service language), D-WIN-3 (service account), D-WIN-4 (start type), D-WIN-5 (pipe encoding) — all presented, none silently resolved (§11.4.66).